EXPERIMENT-10

10. Scenario: You are working on a data visualization project and need to create basic plots using Matplotlib. You have a dataset containing the monthly sales data for a company, including the month and corresponding sales values. Your task is to develop a Python program that generates line plots and bar plots to visualize the sales data.

Question:

1. How would you develop a Python program to create a line plot of the monthly sales data?

2: How would you develop a Python program to create a bar plot of the monthly sales data?

Aim

To create a line plot representing the monthly sales data of a company using Matplotlib.

Algorithm

1. Import the Matplotlib library.
2. Define a list of months and their corresponding sales values.
3. Use plt.plot() to create the line plot.
4. Add a title and label the X-axis and Y-axis.
5. Display grid lines for better readability.
6. Use plt.show() to display the plot.

Code

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [10000, 15000, 12000, 18000, 20000, 25000]
```

```
plt.plot(months, sales, marker='o')

plt.title("Monthly Sales Line Plot")

plt.xlabel("Months")

plt.ylabel("Sales (₹)")

plt.grid(True)

plt.show()
```

Input

```
Months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

Sales = [10000, 15000, 12000, 18000, 20000, 25000]
```

Output

A line plot is displayed showing monthly sales from January to June.

- X-axis: Months
- Y-axis: Sales in rupees
- Sales increase overall from ₹10,000 in January to ₹25,000 in June.
- Sales decrease from February to March.



Result

Thus, the monthly sales data was visualized successfully using a line plot in Matplotlib.

Question 10(b): Bar Plot of Monthly Sales

Aim

To create a bar plot representing the monthly sales data of a company using Matplotlib.

Algorithm

1. Import the Matplotlib library.
2. Define a list of months and their corresponding sales values.
3. Use plt.bar() to create the bar plot.
4. Add a title and label the X-axis and Y-axis.
5. Display the plot using plt.show().

Code

```
import matplotlib.pyplot as plt
```

```
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [10000, 15000, 12000, 18000, 20000, 25000]

plt.bar(months, sales)

plt.title("Monthly Sales Bar Plot")

plt.xlabel("Months")

plt.ylabel("Sales (₹)")

plt.show()
```

Input

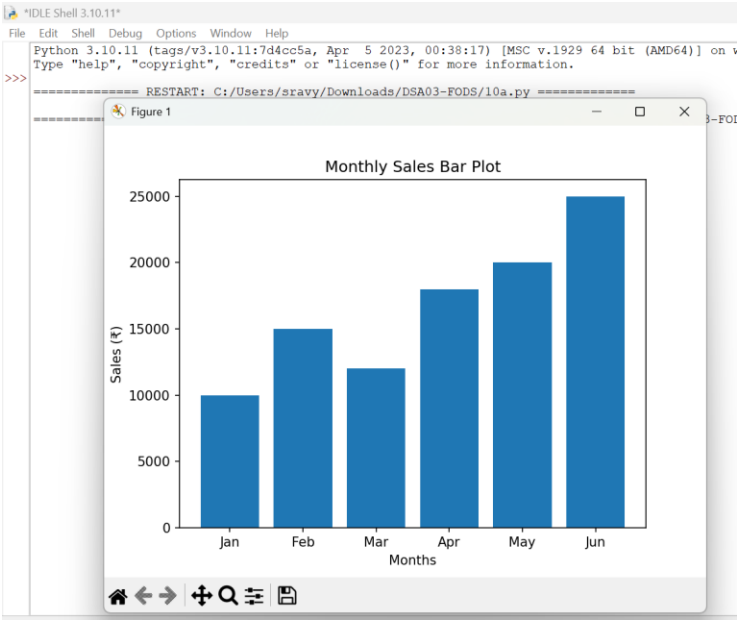
Months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

Sales = [10000, 15000, 12000, 18000, 20000, 25000]

Output

A bar plot is displayed with six bars representing the monthly sales from January to June.

- X-axis: Months
- Y-axis: Sales in rupees
- January has sales of ₹10,000.
- June has the highest sales of ₹25,000.



Result

Thus, the monthly sales data was visualized successfully using a bar plot in Matplotlib.